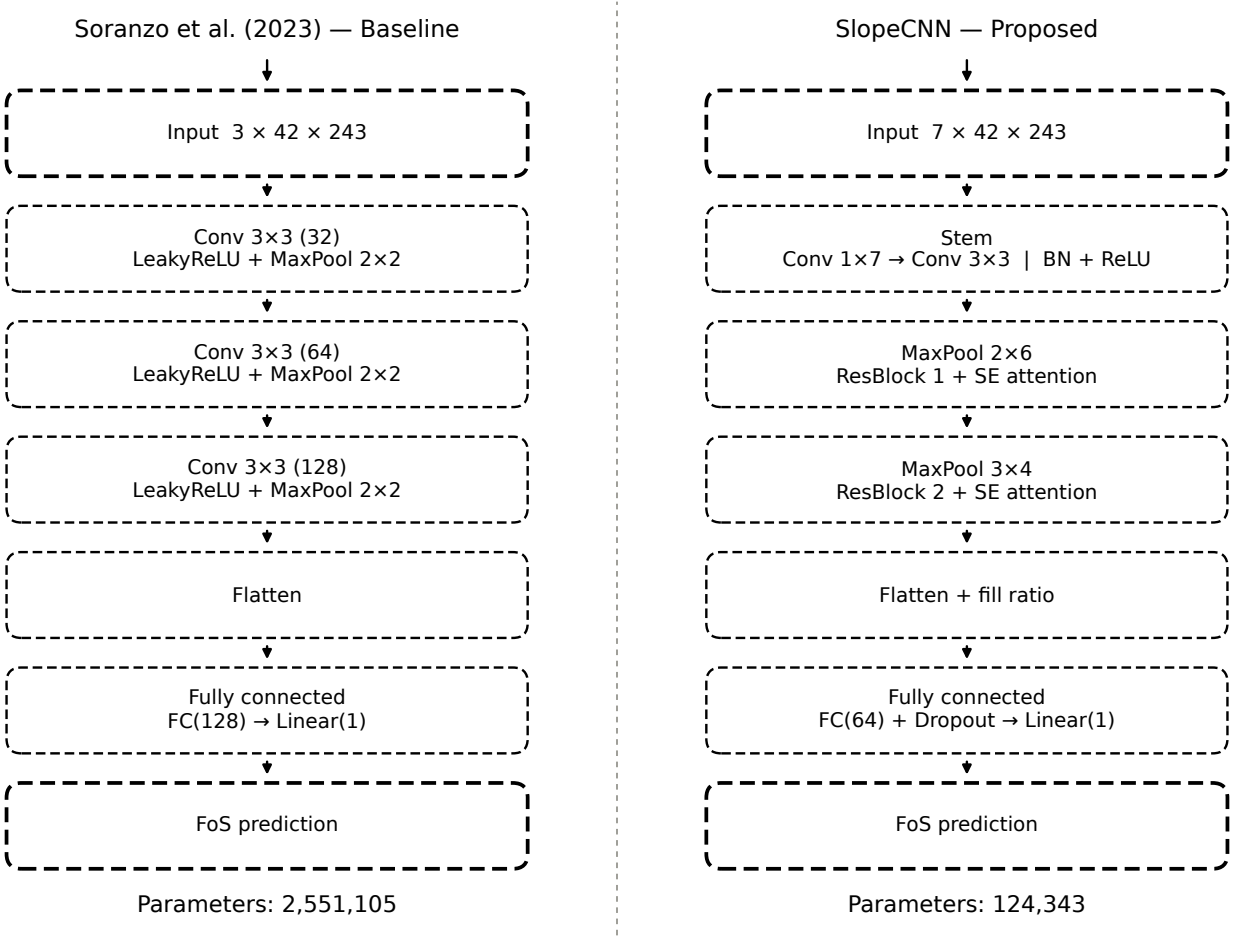
Soranzo et al. (2023) — Baseline
Input 3 × 42 × 243
Conv 3×3 (32)
LeakyReLU + MaxPool 2×2
Conv 3×3 (64)
LeakyReLU + MaxPool 2×2
Conv 3×3 (128)
LeakyReLU + MaxPool 2×2
Flatten
Fully connected
FC(128) → Linear(1)
FoS prediction
Parameters: 2,551,105
SlopeCNN — Proposed
Input 7 × 42 × 243
Stem
Conv 1×7 → Conv 3×3 | BN + ReLU
MaxPool 2×6
ResBlock 1 + SE attention
MaxPool 3×4
ResBlock 2 + SE attention
Flatten + fill ratio
Fully connected
FC(64) + Dropout → Linear(1)
FoS prediction
Parameters: 124,343